

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and Infrastructure Engineering) SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and Infrastructure Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCIC604 - RAILWAY ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Which of the following is NOT a type of track gauge used in India? a) Broad Gauge b) Narrow Gauge c) Standard Gauge d) Ultra Gauge
- 2 What is the main objective of rail alignment? a) To reduce speed b) To ensure smooth operation of trains c) To reduce environmental impact d) To minimize construction costs
- 3 Which of the following factors is crucial for the alignment of a railway line? a) Terrain and gradient b) Train frequency c) The number of stations d) The type of train used
- 4 What is the ideal track gradient in railway line alignment? a) 1-3% b) 5-10% c) 15-20% d) 0-1%
- 5 What is the standard spacing for railway sleepers? a) 0.5m b) 0.6m c) 0.8m d) 1.0m
- 6 What is the minimum depth of ballast recommended under the track? a) 100mm b) 150mm c) 200mm d) 300mm
- 7 What is the primary characteristic of a good ballast? a) High thermal conductivity b) High compressive strength c) Low water absorption d) Low cost
- 8 Which of the following is a basic requirement for ideal alignment of railway tracks? a) Minimal curves and gradients b) Maximum curvature c) Maximum gradients d) Irregular alignment
- 9 What is the primary cause of resistance to traction on curves? a) Curvature and centrifugal force b) Track gradient c) Track modulus d) Track defects
- 10 What is the effect of track wear on resistance to traction? a) It reduces resistance b) It increases resistance c) It has no effect on resistance d) It increases train speed
- 11 Which of the following is a fixed signal type used in railway signaling? a) Semaphore signal b) Radio signal c) Infrared signal d) Satellite signal
- 12 Which type of track layout uses two sets of tracks crossing each other at angles? a) Turnout b) Crossover c) Crossing d) Slip track

Q2. Solve the following.

- A) Draw cross-section of a single broad gauge track in cutting and embankment.

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- B) Explain importance of railways with any six points 6
- Q3. Solve the following.
- A) Explain the drawings required for Railway track project 6
- B) Differentiate between Ruling gradient and Pusher gradient 6
- Q4. Solve Any Two of the following.
- A) Determine the maximum speed of a train on a B.G. track having a curvature of 3 degree and cant of 12 cm. Assume allowable cant deficiency as 76 mm. 6
- B) What is meant by sleeper density? Explain how to find sleeper density 6
- C) What are the specification for track ballast? 6
- Q5. Solve Any Two of the following.
- A) Classify signals and explain any one of them with sketch. 6
- B) Write down the expressions for length of transition according to centrifugal force and super elevation. 6
- C) Derive an expression for widening of gauge on curves 6
- Q6. Solve Any Two of the following.
- A) Draw labeled diagrams of diamond, scissors and triangle crossing. 6
- B) Explain causes and measurement of creep of rail. 6
- C) compare mechanical and electrical signaling system with 5 points each 6

*** End ***